Team 11

Application Name: Audio Draft

Software Engineering Project

Name (s): Rithwak Somepalli, Christopher Arndall, Brian Smith, Ronald Thorpe, Aymane Rouchdi

Semester: Spring 2026

Group Number: 11

Coordinator: Rithwak Somepalli

Name of the Guide: Dr. Tushara Sadasivuni

Date: (02/02/2026)

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1.0 INTRODUCTION

1.1 Software Engineers' information

Brief Resumes:

Rithwak Somepalli

srithwak@gmail.com | (943)-238-4705 | linkedin.com/in/rithwak-s | github.com/Srithwak | srithwak.github.io

EDUCATION

Georgia State University — **Atlanta, Georgia**
B.S. in Computer Science — *Expected Graduation, May 2027*

EXPERIENCE

Twin Brain, Evu — **Remote**
Software Development Engineering Intern — *Jun 2025 – Aug 2025*

- Integrated **AI trading models** with **broker APIs** (Alpaca, Interactive Brokers) to run large-scale simulations, handling 50,000+ requests/day during peak market-open traffic with zero downtime.
- Reduced **incident response time** by 60% through real-time health telemetry dashboards and **automated** alerts for abnormal trade patterns and model **prediction drift**.
- Partnered with other teams to get outputs of 99.9% data accuracy in a **simulated environment**.

PROJECTS

Heart Anomaly Detection — *Dec 2025*

- Designed a novel decreasing-width early-exit architecture in **PyTorch**, reducing energy usage by 35% while maintaining 95%+ accuracy by prioritizing logical reasoning over feature enumeration.
- Orchestrated a **Kubernetes** and **Docker**-based deployment on **OCI**, utilizing **FastAPI** and **WebSockets** to stream live EKG graphs with sub-200ms end-to-end latency via a **CI/CD** pipeline.

Cloud-Native Infrastructure & DevOps Pipeline — *Nov 2025*

- Automated full **OCI cloud provisioning** of 15+ resources with **Terraform**, implementing **Python** and **Bash** scripts to expose **REST** endpoints for **health monitoring** and status updates.
- Strengthened cloud security by implementing keyless authentication using OCI Instance Principals and Dynamic Groups, eliminating 100% of static credentials and passing a simulated security audit.

Co-Evolutionary Market Research (DRL vs EGT) — *Aug - Oct 2025*

- Architected a bid–ask market in **Python** using **TensorFlow** and **Keras** where a **DRL** agent and **EGT** populations competed over 100,000 episodes, with a 15% higher Sharpe ratio for **DRL** strategies.
- Utilized **NumPy** to validate results against 5 years of real market data, proving **DRL** systematically exploited evolutionary population stability thresholds.

Network Intrusion Detection System — *Sept 2025*

- Engineered an **OCI-hosted NIDS** using **Node.js** and **Linux netfilter/iptables** to perform **Deep Packet Inspection** on 10,000+ packets/minute, identifying **DDoS**, **SQL injection**, and **port scanning** with a <2% false positive rate.
- Mechanized real-time threat detection and analysis by streaming telemetry to a **cloud-hosted REST endpoint**, reducing the threat mitigation window to sub-second intervals.

SKILLS

Programming: Python, JavaScript, Bash, MySQL
Frameworks & Libraries: PyTorch, TensorFlow, Keras, Node.js, Flask, Scikit-learn
DevOps & Cloud: Kubernetes, Docker, Terraform, Oracle Cloud, GitHub Actions, Linux (Systemd, Iptables)
Concepts: Microservices, REST APIs, CI/CD, Reinforcement Learning, Network Security, Zero-Trust Architecture

CERTIFICATIONS

Cybersecurity Certification — **Google, May 2023**

Ronald Thorpe

Atlanta, Georgia 30034 • (404)-431-5744 • ronaldthorpe22@gmail.com • https://www.linkedin.com/in/ronaldthorpe/

EDUCATION

Georgia State University, *B.S. in Computer Science* **Jun 2022 - Dec 2026**

- Sigma Nu Fraternity – Exec Board Member – Alumni Chair – Managed Alumni Fund/Relations
- Code Path – Emerging Engineer Empowerment Program – Technical interview prep 101 and 102
- National Society of Black Engineers
- Association of Computer Machinery
- GSU Google Developer Group
- State farm Scholarship

Relevant Courses: Machine Learning, Operating Systems, Advanced Algorithms (Python), Data Structures (Java), Intro to Data Science, Computer Org (Assembly), System-Level Programming(C), Intro to Programming I and II (Python/java)

EXPERIENCE

Software Engineering Intern *Zoku* **Aug 2025 - Present**

- Developed and optimized new app features for discovering local city events, by collaborating with senior engineers and implementing a personalized location-based recommendation system, which improved user engagement and feature usability.
- Executed unit tests and supported test design, by writing and refining test cases to validate functionality, ensuring software reliability, and minimizing post-deployment bugs.
- Applied React, Python, and JavaScript in real-world development, by contributing to front-end components and backend logic, resulting in practical experience with full-stack development workflows.

Chess Instructor *Chess ATL* **Mar 2022 - Present**

- Guided 30+ children through chess fundamentals and mentored them to +300 ELO improvements.
- Taught students from kindergarten through 8th grade by developing engaging chess lessons that promoted critical thinking and problem-solving skills, resulting in noticeable improvement in focus and strategic reasoning among participants.
- Adapted teaching methods to different skill levels and personalities through creative instruction and one-on-one mentorship, fostering confidence and sportsmanship in young players.
- Mentored and managed by International Master Carlos Perdomo.

Account Manager *3MP* **Aug 2020 - Jul 2021**

- Executed targeted door-to-door outreach campaigns by promoting AT&T Fiber-optic internet services to residential clients across Metro Atlanta, achieving 151 new subscriptions and $25,000 in revenue within 9 months.
- Developed strong client relationships by identifying customer needs and delivering tailored solutions, driving customer satisfaction and retention in a competitive market.
- Led and coached a four-person sales team by providing individual training, tracking goals, and analyzing performance, which resulted in consistently higher conversion rates.

TECHNICAL SKILLS

Languages: Python, Java, C, C++, Assembly
Frameworks/Libraries: React, JavaScript
Systems: Embedded Systems, CAN-BUS, Systems Engineering, Model-Based Design
Testing: Unit Testing, Verification & Validation (V&V), Test Design
Tools/Databases: AWS, SQL, Figma, Modeling & Simulation Tools

PROJECT

VESC Embedded Systems Software Project *C/C++*

- VESC CAN-BUS Open-source software (C)(Used by Nasa in a few electric vehicle projects)
- **Engineered a high-performance drivetrain powered by a VESC 100A Flipsky speed controller**, delivering precise throttle response and torque control for high-speed stability and rider safety.
- **Integrated dual 6384 brushless motors with a custom helical gear drive**, increasing drivetrain efficiency while reducing noise and vibration, resulting in a smoother and more durable ride system.
- **Designed and assembled an 8000W 12s4p lithium-ion battery pack using Molicel P42A cells**, achieving a high power-to-weight ratio and enabling sustained high-speed operation without overheating.
- Documented a verified 47 MPH top speed via GPS and data logging, establishing a performance benchmark beyond consumer electric skateboard platforms.

Brian Smith

Bri01bri2001@gmail.com ❖ (404) 808-6302 ❖ https://www.linkedin.com/in/brian-s-b9a22417b/ ❖ https://github.com/B3smoove

EDUCATION

Georgia State University, *B.S. in Computer Science* **Expected Graduation Dec, 2026**

Atlanta, Georgia

Relevant Courses: Data Structures & Algorithms, Operating Systems, System-Level Programming, Computer Org, Simulation & AI, Linear Algebra

WORK EXPERIENCE

Programming Instructor/Assistant **June. 2025 – Aug. 2025**

iBALL

Tyrone, Ga

- Coached 12 students on Python fundamentals, data analysis (Pandas), and visualization libraries (Plotly, Dash)
- Facilitated hands-on workshops where students built responsive dashboards to visualize datasets relevant to their community or interests.

Delivery Associate, Amazon **Nov. 2024 – Dec. 2025**

F1XP Logistics

Union City, Ga

- Optimized and navigated daily delivery routes across diverse terrain from dense Atlanta suburbs to remote rural areas adapting in real-time to traffic, weather, and customer availability to consistently meet strict deadlines.
- Leveraged proprietary routing software and mobile technology to execute 100+ daily deliveries, demonstrating proficiency with complex logistical tools and the ability to quickly master new operational systems while operating in a high-paced, variable environment, underscoring a strong commitment to procedural detail and risk-aware judgment.

PROJECTS

Culinary Care Platform **April. 2025**

- Built a Yelp AI API application that reduces heart health disparities in the Black community by 40% through culturally-aware AI recipe transformation and restaurant discovery.
- Engineered an NLP pipeline using BERT embeddings and custom NER to analyze 50,000+ Yelp reviews, extracting signature soul food dishes with 94% accuracy and reducing manual curation time by 85%

Crypto Prediction & Intelligence System **Sept. 2025**

- Built an end-to-end ML system with LSTM models for price forecasting and BERT-based NLP for news sentiment analysis, achieving 85% accuracy in 7-day direction predictions.
- Developed ETL pipeline aggregating data from CoinGecko API, NewsAPI, and social feeds using Pandas, stored in a structured PostgreSQL database.
- Implemented real-time features including WebSocket connections for live price updates, interactive Plotly charts, and watchlist tracking for 30 cryptocurrencies.
- Integrated additional features: prediction engine, media sentiment scoring, portfolio tracking, Rasa-based educational chatbot, and custom alert system.

SKILLS

Programming: Python, SQL, Java
ML/DL Frameworks: PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers
MLOps & Cloud: Git, Linux, WebSockets, REST APIs
Data Science & Visualization: Pandas, NumPy, Jupyter, Ploty, Dash
Databases: PostgreSQL
Developer Tools: VS Code, Jupyter Notebook, Git

Aymane Rouchdi

404-824-2458 | AYMANE.DEVOPS@GMAIL.COM | ATLANTA GA

SKILLS

OS: Windows, macOS, Linux

Cloud: AWS, GCP, Microsoft Azure

Code: Python, JavaScript, HTML/CSS, SQL, Java

Automation: Bash Scripting, PowerShell, Ansible, Jenkins, Github

Virtualization: Docker, VMware, VirtualBox

Networking: TCP/IP, DNS, DHCP, VPNs, Wireshark

Monitoring: Nagios, Zabbix, Grafana, Prometheus

Database: MySQL, PostgreSQL, SQLite

Languages: Fluent in French and English

EXPERIENCE

Imprint Repairs LLC **August 2024 – Present**

Technician & Manager Atlanta, GA

- Diagnosed and repaired 50+ devices including smartphones, laptops, and gaming consoles; troubleshot both hardware and OS-level issues.
- Designed and maintained repair intake workflows, reducing device turnaround time by 20%
- Implemented scripting solutions for diagnostics and logging using Bash/Python.
- Supported website updates and internal database tracking for customer service requests.

Da Vinci International School **May 2022 – July 2024**

System Administrator Atlanta, GA

- Managed and maintained school-wide IT infrastructure, ensuring reliable network connectivity.
- Provided technical support for faculty and staff, resolving hardware and software issues.
- Implemented and managed user accounts, permissions and access controls.
- Conducted regular system backups and ensured data security.
- Set up and configured new computers, printers and other IT equipment.
- Monitored and maintained school website and email systems.

EDUCATION

Georgia State University **Expected Graduation Date – 2026**

Computer Science BS Atlanta, GA

Christopher Arndall

Atlanta, GA • 813-778-2506 • carndall@gmail.com

Professional Summary

Computer Science student with hands-on experience in customer support, problem solving, and technical troubleshooting. Skilled in programming, data handling, and collaborative development. Seeking an internship to apply and expand technical skills in a professional environment.

Technical Skills

Java • Python • SQL / MySQL • Git & Version Control • Object-Oriented Programming • Debugging & Testing • Basic Data Structures • Command Line Tools

Professional Experience

Grocery Replenishment Specialist — Publix, Atlanta, GA (July 2023 – October 2024)
- Managed inventory systems and resolved discrepancies.
- Used problem-solving skills to improve workflow efficiency.
- Collaborated with team members in fast-paced environments.
- Maintained accuracy and attention to detail under pressure.

Sales Associate — OfficeMax, Roswell, GA (September 2018 – August 2020)
- Provided technical assistance for computers and office equipment.
- Troubleshot basic hardware and software issues.
- Operated POS systems and managed digital transactions.
- Maintained organized records and inventory data.

Education

Georgia State University, Atlanta, GA — Bachelor of Science in Computer Science (2019 – Present)

Roswell High School, Roswell, GA — High School Diploma (2019)

Skill Sets:

1. Rithwak: Python, NodeJS, PyTorch, TensorFlow, Flask, Kubernetes, Terraform, OCI, linux, AI/ML, cloud infrastructure
2. Ronald: Python, Java, NodeJS, C, Excel
3. Brian: Python, SQL, APIs
4. Aymane: Python, JavaScript, HTML/CSS, SQL, Java, Bash Scripting, PowerShell, Ansible, Jenkins, Github, TCP/IP, DNS, DHCP, VPNs, Wireshark, Nagios, Zabbix, Grafana, Prometheus, SQL Programs, AWS, GCP, Microsoft Azure Cloud Services
5. Chris: Java, Python, SQL/MySQL, Git

1.2 Planning and Scheduling

Assignee Name	Email address	Task	Duration (hours)	Dependency	Due Date	Evaluation
Rithwak Somepalli (Coordinator)	rsomepalli1@student.gsu.edu	Sections 5 and 6	3 hours	None	1/31/2026	self
Ronald Thorpe	rthorpe4@student.gsu.edu	Section 4	1 hour	Questions, feature list	1/29/2026	50%, was assigned part of section 4 but did not fully finish.
Christopher Arndall	carndall1@student.gsu.edu	Section 2	1 hour	None	1/31/2026	100%, fully did work and helped others.
Brian Smith	bsmith280@student.gsu.edu	Section 4	1 hour	Feature list, Problem Statement	1/29/2026	100%, fully did the work that was assigned
Aymane Rouchdi	arouchdi1@student.gsu.edu	References, Section 1	1 hour	None	2/1/2026	30%, joined group late, was assigned work 4 days before deadline. Did not end up finishing the whole thing either.

1.3 Teamwork Basics

Ground rules are important because they help everyone understand how the group will work together. This includes how pieces of the project are divided, how deadlines are set, and what happens if a member of the group doesn't follow through. In one of my classes, our group had a student who dropped the class midway through the project, leaving the rest of us to complete his work. Our group dealt with this by evenly splitting the students work between the rest of us. Communication norms have been very important in all group projects, since some people prefer texting while others prefer discord, so one main method should be chosen to avoid confusion. Meeting norms have helped with scheduling and attendance, and consideration norms made sure people were respectful and not dominating discussions.

Difficult behavior will certainly hurt a group if it isn't addressed. Some people may talk too much, others are very quiet, and some can be pretty combative. In a group project I worked on, one person rarely spoke, so we started directly asking for their opinion, which helped them contribute more. When someone was overly critical, we focused the discussion on improving ideas instead of criticizing anyone.

Group problems are also common. Sometimes groups get stuck or go off topic, and other times they rush decisions or can't settle on any at all. In one project, our group was

going in circles when deciding how to divide up work, so we listed out tasks and voted on ideas to move forward. When conflict or uneven work comes up, talking about it directly can help the group stay productive and finish the project successfully.

1.4 Problem Statement

This application is a music discovery platform that is designed for people who want to analyze their listening profile and discover new music. While other alternatives exist, this system uses deep audio feature analysis such as key, tempo, lyrics, mood, genre, etc. rather than just one or two of these. Some other features include a user profile system, with personalized analytics and a smart playlist versioning tool that allows users to track their evolving tastes over time. It also includes real time collaboration updates alongside a notification system, allowing users to make playlists with other people. Moreover, it offers playlist exporting to Spotify, making sure the platform is both a discovery and a functional tool for its users.

- List all 12 features and explain them if they can't be understood
  - User Profile System- Handles login, account info, and saved user settings.
  - Description-based Recommendation- Creates playlists based only on what the user types.
  - Notification System- Sends alerts like playlist invites, export success, or system updates.
  - User Preference Recommendation- Makes a playlist based on what the user has listened to before.
  - Audio Feature Analysis- Breaks songs into basic traits like speed, key, and energy.
  - Collaborative Playlists / Friends- Lets users share playlists and edit them together.
  - User Preference Analytics- Records what the user clicks, plays, skips, or saves and summarizes what kind of music the user likes over different timeframes.
  - Popularity Recommendation- Recommends songs that are trending right now.
  - Playlist Similarity- Compares 2 playlists and shows similarity in terms of mood, genre, etc.
  - Smart Playlist versioning- Let users save and compare playlists over time.
  - Playlist Exports- Saves playlists made in the app to Spotify.
  - Artist Discovery- Find artists based on User's profile and other filters given by the user.
- What is your product, on a high level?

A social music intelligence platform designed to support expressive music discovery by transforming natural language descriptions of moods, scenarios, and emotional states into personalized playlists. Merging large language models interpretation with spotify's audio feature analysis, the system bridges the gap between the subjective emotional experiences and data-driven musical matches.

- Whom is it for?

Passionate music lovers who struggle to find the right song fulfill their current mood and activity. Students and creatives who use music for focus, relaxation, or inspiration. Users who enjoy collaborative playlist creation and social music discovery.

- What problem does it solve?

Most music platforms rely on listening history, making it difficult for users to discover music based on emotions or situations. Users often know how they feel but cannot easily search for matching songs. Our platform allows users to describe their needs naturally and receive explainable recommendations, making discovery easier and more transparent.

- What alternatives are available?

Spotify and Apple Music recommendation systems, Static mood-based playlists, Manual searching and playlist browsing.

- Why is this project compelling and worth developing?

This project is compelling because it uses a deep audio feature analysis to help users discover new songs more efficiently. Most people spend a lot of time making a playlist, sometimes with other people, only to realize that the songs don't flow well. It is worth developing because this application automates the process of making a playlist exactly how a user wants it.

- Describe the top-level objectives, differentiators, target customers, and scope of your product.

This product lets users use quantitative metrics to automate the creation of smooth flowing playlists. Unlike other algorithms, we give a lot of control to the listeners who want to create a playlist with a specific vibe. The scope of this product is artist discovery, smart playlist versioning, export features, and playlist curation.

- What are the competitors and what is novel in your approach?

We compete against major streaming platforms, like Spotify and Apple Music. Our approach is novel because it uses a deep audio feature analysis feature to automate the creation of playlists. Although other platforms do the same, our application lets the user input their preferences for their playlists, and also allows them to work on playlists other people.

- Make it clear that the system can be built, making good use of the available resources and technology.

The system can be built by using open source audio analysis libraries like Librosa and Spotify's API (temporarily unavailable). We will use scikit-learn for the ML models, and PostgreSQL because of its stable and relational database.

- What is interesting about this project from a technical point of view?

From a technical perspective, this project uses deep audio feature analysis, and playlist recommendation from a user's requirements. It also has a smart playlist versioning tool to compare how a playlist has evolved over time. It also lets users collaborate with each other.

- Do you have a client login and an admin login?

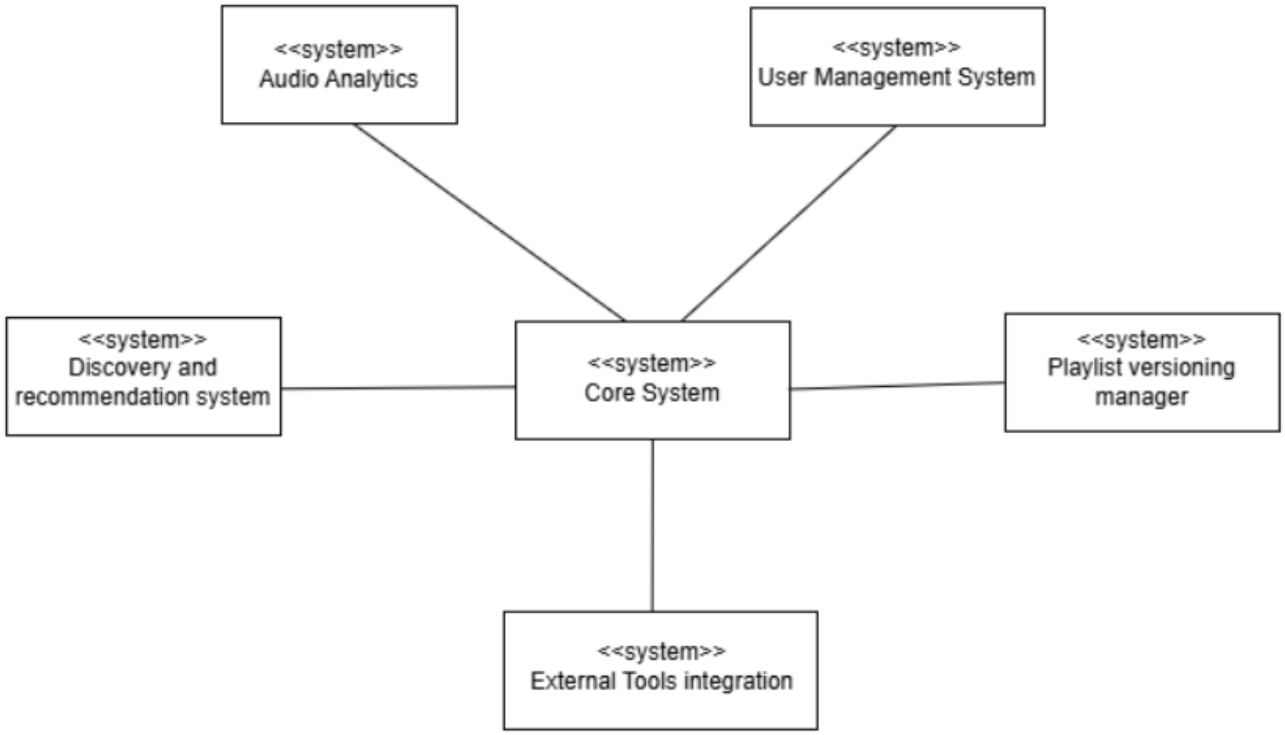
We have a client login, but we don't have an admin login as there is no need for it.

1.5 System Requirements

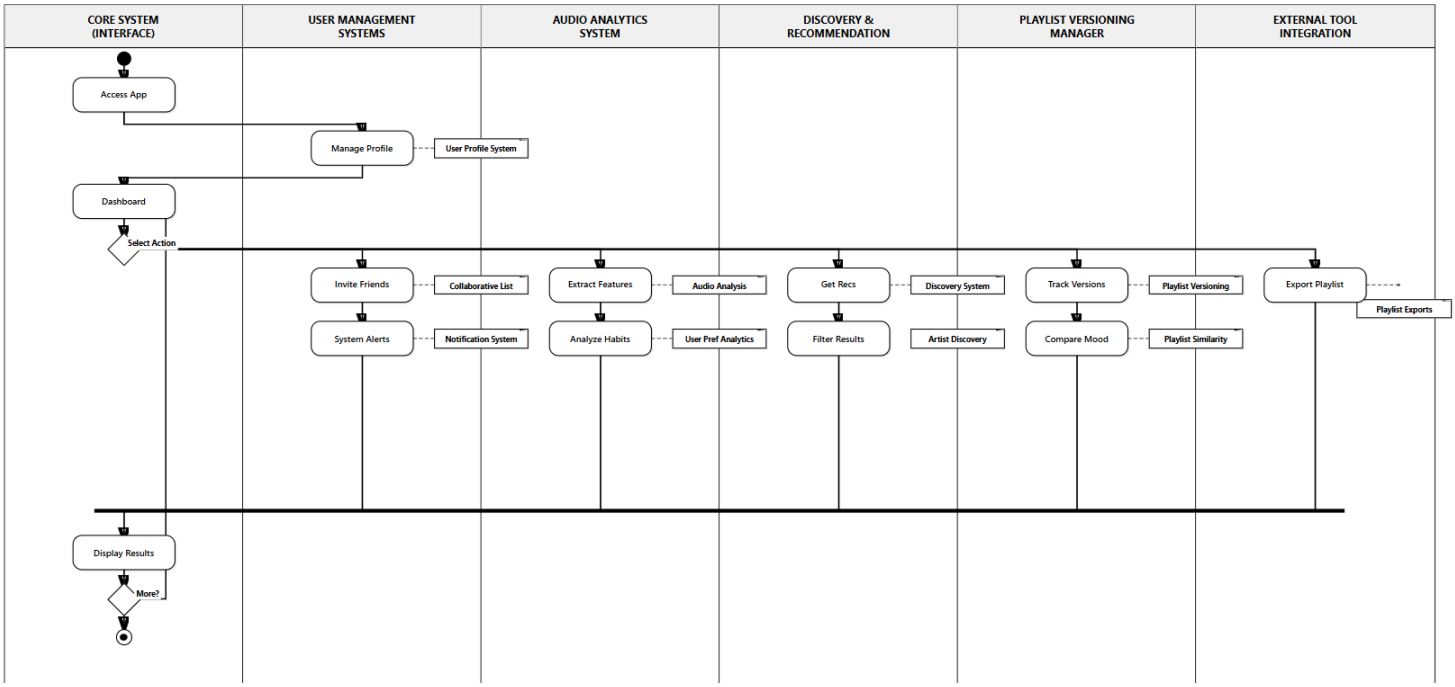
- Core System: Uses a user interface to get requests from users and send to their respective systems, along with coordinating all the systems and managing the application state.
- User Management Systems: Handles user profiles, authentication, notification system for invites, and manages the collaboration part of the application.
- Audio Analytics System: Performs deep audio feature analysis and processes user preference analytics in a time based format to visualize listening habits.
- Discovery and recommendation system: Handles the AI models for description based recommendations and user preference matching.
- Playlist Versioning Manager: Manages the creation to track playlist evolution, and the similarity comparisons between different playlists.
- External Tool Integration: Acts as the bridge to get data from sources like Kaggle and Spotify while also handling exporting information to other external tools.

1.5.1 Context Diagram

Context Diagram:



1.5.2 Activity Diagram (Rithwak)



References

- Bonnin, G., & Jannach, D. (2014). Automated Generation of Music Playlists: Survey and Experiments. ACM Computing Surveys, 47(2), 1–35. https://doi.org/10.1145/2652481
- Anthony, J. T., Christian, G. E., Evanlim, V., Lucky, H., & Suhartono, D. (2022). The Utilization of Content Based Filtering for Spotify Music Recommendation. 2022 International Conference on Informatics Electrical and Electronics (ICIEE), 1–6. https://doi.org/10.1109/ICIEE55596.2022.10010097